



# AI FOR OCEAN EXPLORATION: Is An AI Colony Possible In The Deep-Sea?

Artificial intelligence (AI) and autonomous robots are used to speed up the exploration and study of hard-to-reach deep-sea ecosystems. Improvements in AI and robotics in ocean exploration are likely to benefit researchers in multiple areas, especially those handling dangerous tasks. Several giants in the oil and gas industries are integrating AI for innovation and sustainable energy strategies, driving the pace of evolution in the field

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(Credit: www.biometricupdate.com)

**L**ife on Earth probably began in the depths of the ocean. More than 71 per cent of the Earth's surface is covered by oceans and yet 95 per cent of the world's oceans are still unexplored and unseen by humans. The oceans are full of mystery including the Atlantis of Japan, Baltic Sea anomaly, Bermuda Triangle, marine species and others. Average depth of the oceans is around 3700 metres—this leaves room for the mysterious, mythical and mystical things to exist.

It is also true that the oceans have a great potential for military and civil applications, and all these let humans explore the deep-sea. Oceans have a tremendous amount of energy including tidal, wave, ocean current, ocean thermal and osmotic. There is no dearth of developments in the fields of deep-sea mining, research, communication, submarines and so on. This article covers how robots and artificial intelligence (AI)

are used to explore those parts of the oceans that the humans cannot reach.

### Understanding the deep-sea

Benefits of exploring the ocean deep down are plentiful, but to fully explore and understand the deep-sea will take some time. Apart from the benefits of getting oil and gas from oceans, there are possibilities of finding new medicines, understanding climate change and helping researchers identify endangered species. Also, exploring the deep-sea could help humans extract energy resources, understand the seabed and monitor natural disasters like earthquakes and tsunamis. Image capture and analysis is an important step to extract useful information from the oceans.

Researchers in Germany have developed a new standardised workflow for sustainable marine image analysis that is divided into three steps: data acquisition, data cura-